

Competitive Programming

Section 24 Editorial: Grid DP

Competitive Programming Course

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Table of contents

1	Editorial overview	3
2	A. Safe Paths	4
2.1	Restatement	4
2.2	Constraints and traps	4
2.3	Observations	4
2.4	Sample walkthrough	4
2.5	Algorithm	5
2.6	Correctness proof	5
2.7	Complexity	5
2.8	Full C++ solution	5
2.9	Full Python solution	6
2.10	Common wrong answers	7
2.11	Tests that matter	7
3	B. Lowest Toll	8
3.1	Restatement	8
3.2	Constraints and traps	8
3.3	Observations	8
3.4	Sample walkthrough	8
3.5	Algorithm	9
3.6	Correctness proof	9
3.7	Complexity	9
3.8	Full C++ solution	9
3.9	Full Python solution	10
3.10	Common wrong answers	11
3.11	Tests that matter	11
4	C. Limited Turns	12
4.1	Restatement	12
4.2	Constraints and traps	12
4.3	Observations	12
4.4	Sample walkthrough	12
4.5	Algorithm	13
4.6	Correctness proof	13
4.7	Complexity	13
4.8	Full C++ solution	14
4.9	Full Python solution	15
4.10	Common wrong answers	16
4.11	Tests that matter	16
5	Testing and stress strategy	17
6	What to carry forward	18

1 Editorial overview

Problem	State	Transition
A. Safe Paths	ways to reach cell	sum from top and left
B. Lowest Toll	minimum cost to reach cell	toll + min(top, left)
C. Limited Turns	cell, turns, last direction	continue or turn

All three problems rely on the same row-major computation order. The third problem adds direction because the number of turns cannot be determined from the cell alone.

2 A. Safe Paths

2.1 Restatement

Given an $n \times m$ grid with open and blocked cells, count paths from the top-left to the bottom-right using only right and down moves. Paths may not enter blocked cells. Print the count modulo 1_000_000_007.

2.2 Constraints and traps

n and m can be up to 1000, so an $O(nm)$ DP is intended.

Important traps:

```
Blocked cells contribute zero paths.  
If the start or finish is blocked, the answer is zero.  
The answer can be huge, so take modulo after additions.  
Only right/down moves are allowed.
```

2.3 Observations

Every path reaching (r, c) must enter from:

```
(r-1, c) above  
(r, c-1) left
```

No other cell can be the immediate predecessor under right/down movement.

2.4 Sample walkthrough

Grid:

```
....  
.#..  
....
```

DP table:

```
1 1 1 1  
1 # 1 2  
1 1 2 4
```

Answer: 4.

2.5 Algorithm

```
if start is open:
    dp[0][0] = 1

for r from 0 to n-1:
    for c from 0 to m-1:
        if cell is blocked:
            dp[r][c] = 0
        else:
            add dp[r-1][c] if r > 0
            add dp[r][c-1] if c > 0

print dp[n-1][m-1]
```

2.6 Correctness proof

Lemma. For every open cell (r, c) , every valid path ending at (r, c) has immediate predecessor $(r-1, c)$ or $(r, c-1)$, unless (r, c) is the start.

Proof. The only allowed moves are down and right. Reversing the final move, a path into (r, c) must have come from one row above or one column left. $\square$

Theorem. The algorithm computes the number of valid paths to every cell.

Proof. Induct over row-major order. The start cell is initialized correctly. For any blocked cell, no valid path can stand there, so zero is correct. For an open non-start cell, the lemma partitions all paths by their final predecessor. Those sets are disjoint, and the predecessor values have already been computed in row-major order. The transition adds exactly their counts. $\square$

2.7 Complexity

```
Time:  $O(nm)$ 
Memory:  $O(nm)$ 
```

2.8 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
```

```

const long long MOD = 1000000007LL;
int n, m;
if (!(cin >> n >> m)) return 0;
vector<string> grid(n);
for (string &row : grid) cin >> row;
vector<vector<long long>> dp(n, vector<long long>(m, 0));
if (grid[0][0] == '.') dp[0][0] = 1;
for (int r = 0; r < n; r++) {
    for (int c = 0; c < m; c++) {
        if (grid[r][c] == '#') {
            dp[r][c] = 0;
            continue;
        }
        if (r) dp[r][c] = (dp[r][c] + dp[r - 1][c]) % MOD;
        if (c) dp[r][c] = (dp[r][c] + dp[r][c - 1]) % MOD;
    }
}
cout << dp[n - 1][m - 1] << '\n';
return 0;
}

```

2.9 Full Python solution

```

import sys

MOD = 1_000_000_007

def main() -> None:
    input = sys.stdin.readline
    first = input().split()
    if not first:
        return
    n, m = map(int, first)
    grid = [input().strip() for _ in range(n)]
    dp = [[0] * m for _ in range(n)]
    if grid[0][0] == ".":
        dp[0][0] = 1
    for r in range(n):
        for c in range(m):
            if grid[r][c] == "#":
                dp[r][c] = 0
                continue
            if r:
                dp[r][c] = (dp[r][c] + dp[r - 1][c]) % MOD
            if c:
                dp[r][c] = (dp[r][c] + dp[r][c - 1]) % MOD

```

```
print(dp[n - 1][m - 1])

if __name__ == "__main__":
    main()
```

2.10 Common wrong answers

Not zeroing blocked cells.
Letting paths pass through a blocked start or finish.
Forgetting modulo.
Using diagonal or upward moves.

2.11 Tests that matter

1x1 open grid.
Blocked finish.
A full open grid with many paths.
A wall that separates start from finish.

3 B. Lowest Toll

3.1 Restatement

Each grid cell has a nonnegative toll. Starting at the top-left and moving only right/down, reach the bottom-right with minimum total toll, including both start and finish cells.

3.2 Constraints and traps

$n, m \leq 1000$, so $O(nm)$ is intended.

Important traps:

```
The start cell's toll is included.  
The finish cell's toll is included.  
Tolls can be large, so use long long.  
Only the previous row is needed for memory.
```

3.3 Observations

The best path to (r, c) must come from the better of:

```
best path to  $(r-1, c)$   
best path to  $(r, c-1)$ 
```

Then add the current toll.

3.4 Sample walkthrough

Grid:

```
5 9 1  
4 2 7  
6 1 3
```

DP:

```
5 14 15  
9 11 18  
15 12 15
```

Answer: 15.

3.5 Algorithm

```
prev = INF row

for each row r:
    cur = INF row
    for each column c:
        if start:
            cur[c] = toll
        else:
            cur[c] = toll + min(prev[c], cur[c-1])
    prev = cur

print prev[m-1]
```

3.6 Correctness proof

Lemma. Any valid path to (r, c) must enter from above or left, except for the start.

Proof. The only moves are right and down. $\square$

Theorem. The algorithm computes the minimum cost to each processed cell.

Proof. Induct over row-major order. The start is initialized to its own toll. For any other cell, by the lemma, the final predecessor is above or left. By induction, `prev[c]` and `cur[c-1]` contain the optimal costs to those predecessors. Taking their minimum and adding the current toll gives the best possible path to (r, c) . $\square$

3.7 Complexity

```
Time:  $O(nm)$ 
Memory:  $O(m)$ 
```

3.8 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int n, m;
    if (!(cin >> n >> m)) return 0;
    const long long INF = (1LL << 62);
```

```

vector<long long> prev(m, INF), cur(m, INF);
for (int r = 0; r < n; r++) {
    fill(cur.begin(), cur.end(), INF);
    for (int c = 0; c < m; c++) {
        long long x;
        cin >> x;
        if (r == 0 && c == 0) {
            cur[c] = x;
        } else {
            long long best = prev[c];
            if (c) best = min(best, cur[c - 1]);
            cur[c] = best + x;
        }
    }
    swap(prev, cur);
}
cout << prev[m - 1] << '\n';
return 0;
}

```

3.9 Full Python solution

```

import sys

def main() -> None:
    data = list(map(int, sys.stdin.buffer.read().split()))
    if not data:
        return
    it = iter(data)
    n = next(it)
    m = next(it)
    inf = 10**30
    prev = [inf] * m
    for r in range(n):
        cur = [inf] * m
        for c in range(m):
            x = next(it)
            if r == 0 and c == 0:
                cur[c] = x
            else:
                best = prev[c]
                if c:
                    best = min(best, cur[c - 1])
                cur[c] = best + x
        prev = cur
    print(prev[-1])

```

```
if __name__ == "__main__":  
    main()
```

3.10 Common wrong answers

Forgetting to include the starting toll.
Using int for a path of many large tolls.
Using max instead of min after copying a counting recurrence.
Incorrect first row or first column initialization.

3.11 Tests that matter

1x1 grid.
One row.
One column.
Large tolls.
Best route intentionally avoids a locally cheap cell.

4 C. Limited Turns

4.1 Restatement

Count right/down paths through open cells from the top-left to the bottom-right that use at most K turns. A turn is a change between right and down on two consecutive moves. The first move is not a turn.

4.2 Constraints and traps

$n, m \leq 80$ and $K \leq 40$, so a state with cell, turn count, and direction is small enough.

Important traps:

```
dp[r][c] alone is not enough.  
The first move has no previous direction and uses zero turns.  
The 1x1 grid has one empty path.  
Blocked cells have zero states.  
The answer asks for at most  $K$  turns, so sum over  $0..K$ .
```

4.3 Observations

To decide whether entering (r, c) creates a turn, we must know the previous last direction. Therefore:

```
dp[r][c][t][dir]
```

where `dir` is the last move direction.

4.4 Sample walkthrough

For a 3×3 open grid with at most one turn, the only valid paths are:

```
RRDD  
DDRR
```

Every other right/down path changes direction at least twice. Answer: 2.

4.5 Algorithm

```
if start or finish is blocked: print 0
if n = 1 and m = 1: print 1

initialize first right move with 0 turns
initialize first down move with 0 turns

for every cell in row-major order:
    for turns from 0 to K:
        enter from left:
            continue RIGHT
            or turn from DOWN to RIGHT
        enter from above:
            continue DOWN
            or turn from RIGHT to DOWN

answer = sum both directions for turns 0..K at finish
```

4.6 Correctness proof

Lemma 1. Every non-empty path ending at a cell has exactly one last move direction: right or down.

Proof. The allowed moves are right and down. The final move, if it exists, is exactly one of them. $\square$

Lemma 2. The transition counts every valid path ending at (r,c) with exactly t turns and last direction dir once.

Proof. Consider $dir = RIGHT$. Such a path must enter from the left. Before the final move, it either already ended with $RIGHT$, adding no turn, or ended with $DOWN$, adding one turn. The transition includes exactly these two cases. They are disjoint by previous last direction. The $DOWN$ case is symmetric. $\square$

Theorem. The algorithm prints the number of valid paths with at most K turns.

Proof. The initialized first moves correctly represent one-move paths with zero turns. By induction over row-major order and Lemma 2, every DP state has the intended count. Summing the finish states over both last directions and turn counts $0..K$ gives exactly the requested paths. The 1×1 case is handled as the unique empty path. $\square$

4.7 Complexity

```
Time:  $O(nmK)$ 
Memory:  $O(nmK)$ 
```

The direction factor is constant.

4.8 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    const int MOD = 1000000007;
    const int RIGHT = 0, DOWN = 1;
    int n, m, K;
    if (!(cin >> n >> m >> K)) return 0;
    vector<string> grid(n);
    for (string &row : grid) cin >> row;
    if (grid[0][0] == '#' || grid[n - 1][m - 1] == '#') {
        cout << 0 << '\n';
        return 0;
    }
    if (n == 1 && m == 1) {
        cout << 1 << '\n';
        return 0;
    }
    vector dp(n, vector(m, vector(K + 1, array<int, 2>{0, 0})));
    if (m > 1 && grid[0][1] == '.') dp[0][1][0][RIGHT] = 1;
    if (n > 1 && grid[1][0] == '.') dp[1][0][0][DOWN] = 1;
    for (int r = 0; r < n; r++) {
        for (int c = 0; c < m; c++) {
            if (grid[r][c] == '#') continue;
            for (int turns = 0; turns <= K; turns++) {
                if (c > 0 && grid[r][c - 1] == '.') {
                    int val = dp[r][c][turns][RIGHT];
                    val += dp[r][c - 1][turns][RIGHT];
                    if (val >= MOD) val -= MOD;
                    if (turns) {
                        val += dp[r][c - 1][turns - 1][DOWN];
                        if (val >= MOD) val -= MOD;
                    }
                    dp[r][c][turns][RIGHT] = val;
                }
                if (r > 0 && grid[r - 1][c] == '.') {
                    int val = dp[r][c][turns][DOWN];
                    val += dp[r - 1][c][turns][DOWN];
                    if (val >= MOD) val -= MOD;
                    if (turns) {
                        val += dp[r - 1][c][turns - 1][RIGHT];
                        if (val >= MOD) val -= MOD;
                    }
                    dp[r][c][turns][DOWN] = val;
                }
            }
        }
    }
}
```

```

    }
}
}
int ans = 0;
for (int turns = 0; turns <= K; turns++) {
    ans += dp[n - 1][m - 1][turns][RIGHT];
    if (ans >= MOD) ans -= MOD;
    ans += dp[n - 1][m - 1][turns][DOWN];
    if (ans >= MOD) ans -= MOD;
}
cout << ans << '\n';
return 0;
}

```

4.9 Full Python solution

```

import sys

MOD = 1_000_000_007
RIGHT = 0
DOWN = 1

def main() -> None:
    input = sys.stdin.readline
    first = input().split()
    if not first:
        return
    n, m, k = map(int, first)
    grid = [input().strip() for _ in range(n)]
    if grid[0][0] == "#" or grid[n - 1][m - 1] == "#":
        print(0)
        return
    dp = [[[0, 0] for _ in range(k + 1)] for _ in range(m)] for _ in range(n)]
    if m > 1 and grid[0][1] == ".":
        dp[0][1][0][RIGHT] = 1
    if n > 1 and grid[1][0] == ".":
        dp[1][0][0][DOWN] = 1
    if n == 1 and m == 1:
        print(1)
        return
    for r in range(n):
        for c in range(m):
            if grid[r][c] == "#":
                continue
            for turns in range(k + 1):
                if c > 0 and grid[r][c - 1] == ".":

```

```

        dp[r][c][turns][RIGHT] = (
            dp[r][c][turns][RIGHT] + dp[r][c - 1][turns][RIGHT]
        ) % MOD
    if turns:
        dp[r][c][turns][RIGHT] = (
            dp[r][c][turns][RIGHT] + dp[r][c - 1][turns - 1][DOWN]
        ) % MOD
    if r > 0 and grid[r - 1][c] == ".":
        dp[r][c][turns][DOWN] = (
            dp[r][c][turns][DOWN] + dp[r - 1][c][turns][DOWN]
        ) % MOD
    if turns:
        dp[r][c][turns][DOWN] = (
            dp[r][c][turns][DOWN] + dp[r - 1][c][turns - 1][RIGHT]
        ) % MOD

ans = 0
for turns in range(k + 1):
    ans = (ans + dp[n - 1][m - 1][turns][RIGHT] + dp[n - 1][m - 1][turns][DOWN]) % MOD
print(ans)

if __name__ == "__main__":
    main()

```

4.10 Common wrong answers

Counting the first move as a turn.
 Forgetting to sum all turn counts up to K.
 Not storing last direction.
 Allowing paths to leave or enter blocked cells.
 Mishandling the 1x1 grid.

4.11 Tests that matter

1x1 grid.
 2x2 grid with K = 0 -> no path.
 Open 3x3 with K = 1 -> two paths.
 Blocked first right or first down cell.
 Large K should match ordinary path counting.

5 Testing and stress strategy

Run:

```
CP_TARGET=solution python3 sections/24_grid_dp/check.py
```

The random tests cover:

Safe Paths:

random blocked cells and zero-path grids.

Lowest Toll:

one-row, one-column, and random weighted grids.

Limited Turns:

small random grids with different K values and obstacles.

For tiny stress tests:

A: enumerate all right/down move strings for $n, m \leq 7$.

B: enumerate all right/down paths and take the minimum cost.

C: enumerate all right/down paths and count turns directly.

6 What to carry forward

Grid DP is about dependency direction.

```
If movement is acyclic, choose an order and fill a table.  
If a condition depends on history, add exactly the missing state.  
If only the previous row matters, row compression is available.
```

The next DP sections will use the same habits with intervals, trees, and bitmasks, where the correct order is less visually obvious.